

数值分析第七次作业

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1. 考虑等距节点 $x_k = x_0 + kh, 0 \leq k \leq n, h > 0$.

(1) 假设 $f \in C^n[x_0, x_n]$, 求证: 当 $h \ll 1$ 时, Newton 差商

$$f[x_0, x_1, \dots, x_n] \sim \mathcal{O}(1).$$

(2) 假设 f 在 $[x_0, x_n]$ 上为右连续的分片光滑函数, 且存在唯一的第一类间断点 $x^* \in (x_0, x_n)$, 求证: 当 $h \ll 1$ 时, Newton 差商

$$f[x_0, x_1, \dots, x_n] \sim \mathcal{O}(h^{-n}).$$

解答.

(1) 由 Newton 差商的性质可得

$$f[x_0, x_1, \dots, x_n] = \frac{f^{(n)}(\xi)}{n!}, \quad \xi \in (x_0, x_n),$$

又 $h \ll 1$, 故 $\xi = x_0 + o(1)$, 再结合 $f \in C^n[x_0, x_n]$ 可得 $f^{(n)}(\xi) = f^{(n)}(x_0) + o(1)$, 因此

$$f[x_0, x_1, \dots, x_n] = \frac{f^{(n)}(x_0)}{n!} + o(1) \sim \mathcal{O}(1).$$

(2) 设

$$x^* \in (x_m, x_{m+1}], \quad 0 \leq m \leq n-1, \quad \alpha = f(x^* - 0), \quad \beta = f(x^*), \quad \alpha \neq \beta.$$

由 Newton 差商的性质

$$f[x_0, x_1, \dots, x_n] = \sum_{k=0}^n \frac{f(x_k)}{\omega'_{n+1}(x_k)}, \quad \omega_{n+1}(x) := \prod_{k=0}^n (x - x_k),$$

同时直接计算可得

$$\omega'_{n+1}(x_k) = \prod_{j \neq k} (x_k - x_j) = h^n \prod_{j \neq k} (k - j) = (-1)^{n-k} k! (n - k)! h^n = \frac{(-1)^{n-k} n!}{C_n^k} h^n.$$

再结合 $x^* - 0$ 和 x^* 处的 Taylor 展开知

$$\begin{aligned}
& f[x_0, x_1, \dots, x_n] \\
&= \sum_{k=0}^m \frac{f(x_k)}{\omega'_{n+1}(x_k)} + \sum_{k=m+1}^n \frac{f(x_k)}{\omega'_{n+1}(x_k)} \\
&= \frac{1}{n!} h^{-n} \left(\sum_{k=0}^m (-1)^{n-k} C_n^k (\alpha + \mathcal{O}(x_k - x^*)) + \sum_{k=m+1}^n (-1)^{n-k} C_n^k (\beta + \mathcal{O}(x_k - x^*)) \right) \\
&= \frac{1}{n!} h^{-n} \left(\sum_{k=0}^m (-1)^{n-k} C_n^k (\alpha + \mathcal{O}(h)) + \sum_{k=m+1}^n (-1)^{n-k} C_n^k (\beta + \mathcal{O}(h)) \right) \\
&= \frac{1}{n!} h^{-n} \left(\alpha \sum_{k=0}^m (-1)^{n-k} C_n^k + \beta \sum_{k=m+1}^n (-1)^{n-k} C_n^k \right) + \mathcal{O}(h^{-n+1}) \\
&= \frac{1}{n!} h^{-n} \left(\alpha \sum_{k=0}^n (-1)^{n-k} C_n^k + (\beta - \alpha) \sum_{k=m+1}^n (-1)^{n-k} C_n^k \right) + \mathcal{O}(h^{-n+1}) \\
&= (-1)^{n+m+1} C_{n-1}^m \frac{\beta - \alpha}{n!} h^{-n} + \mathcal{O}(h^{-n+1}) \sim \mathcal{O}(h^{-n}),
\end{aligned}$$

这里用到了 $x_k - x^* = \mathcal{O}(h), \forall 0 \leq k \leq n$, 和

$$\sum_{k=0}^n (-1)^{n-k} C_n^k = (1-1)^n = 0,$$

以及

$$\sum_{k=m+1}^n (-1)^{n-k} C_n^k = (-1)^n \sum_{k=m+1}^n [(-1)^k C_{n-1}^k - (-1)^{k+1} C_{n-1}^{k-1}] = (-1)^{n+m+1} C_{n-1}^m \neq 0.$$

□